

W-4. Water Harvesting Pond				
Water Resources Development	Medium	High		
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions									
Measures	In arid and semi-arid locations where, seasonal rainfall is insufficient to support household and agricultural needs, water scarcity is becoming a more serious problem worldwide. Rainwater may be efficiently collected and stored in water harvesting ponds, providing a consistent supply of water for a variety of uses. To preserve the environment and maximize the benefits of these ponds, however, their construction and maintenance require for meticulous planning and conservation measures.								
	Bangladesh's CHT, which are known for their indigenous populations and diverse ecosystems, face several difficulties in managing their water resources. Steep hills and erratic rainfall patterns define the region's distinctive geography, which makes efficient water gathering essential for both household and agricultural use. Rainwater can be collected and stored by water harvesting ponds, giving communities nearby a reliable supply of water. Ensuring a balanced approach to water resource development and ecological preservation by giving conservation top priority in the management of water harvesting ponds will provide a sustainable future for the CHT' inhabitants and environment.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
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Applicable Sites	• Moderate to Steep Slopes (5-15%) – Requires terracing and reinforced embankments. • Low-Lying Catchment Areas – Naturally collects runoff, reducing excavation costs. Deforested or Degraded Land – Helps in soil conservation and groundwater recharge. • Near Agricultural Fields & Villages – Provides easy access for irrigation and household use.								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
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Design options and performance	• Suitable for low-lying areas, allowing natural runoff collection. • Constructed on slopes with reinforced bunds to store water. • Step-like water storage structures ideal for hilly terrain, reducing soil erosion. • Designed for household, livestock, and irrigation use with filtering mechanisms.								
Feasibility criteria	Technical Feasibility • Ensures reduced seepage and minimal erosion.								

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	- Assesses catchment size and runoff potential. - Ensures adequate depth, storage volume, and outlet structures. Environmental Feasibility - Prevents runoff-related degradation. - Enhances groundwater replenishment. - Supports local flora and fauna by creating aquatic habitats. Economic Feasibility - Utilizes locally available materials for excavation and lining. - Improves crop yields, food security, and household water availability. - Encourages community-led management to reduce financial burden. Social Feasibility - Engages local populations in construction and management. - Supports farming, fisheries, and household water supply.			
Operation and maintenance	Routine Maintenance – embankment repair, and vegetation control. Structural Repairs – Ensures bund stability, spillway function, and leak prevention. Seasonal Adjustments – Prepares for monsoon overflow and dry-season water conservation Community Involvement – Establishes local water user groups for monitoring and upkeep.			
Cost	Medium			
Benefits	Benefits are as follows-			
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			

	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	• Water Conservation – Enhances groundwater recharge and reduces wastage. • Climate Resilience – Mitigates drought effects and regulates seasonal water availability. • Ecosystem Protection – Supports wetland restoration and biodiversity conservation.			
Remarks	Selecting the right site is critical, as locations must be geologically stable and hydrologically viable to ensure long-term functionality. The sustainability of these projects depends on regular maintenance, with active community participation playing a key role in their success. Additionally, reservoirs should be integrated into broader watershed management efforts to enhance overall catchment conservation. Collaboration between the government and NGOs is essential, providing policy support, funding, and technical expertise to ensure efficient implementation and long-term viability.			